

```
1 % Part 1c
2 function approx = exp_function(x)
3 tolerance = 1e-6;
4 approx = 0;
5 k = 0;
6
7 while true
8     term = (x^k) / factorial(k);
9     next_approx = approx + term;
10
11     if k > 0
12         err_rel = abs((next_approx - approx) / next_approx);
13         if err_rel < tolerance
14             approx = next_approx;
15             break;
16         end
17     end
18     approx = next_approx;
19     k = k + 1;
20 end
21 end
22
```